

GE Healthcare

Signa 16 Channel RF Electronics Upgrade

Technical Publication



5177002
Rev 3

DAMAGE IN TRANSPORTATION

All packages should be closely examined at time of delivery. If damage is apparent, have notation “**damage in shipment**” written on **all** copies of the freight or express bill **before** delivery is accepted or “signed for” by a General Electric representative or a hospital receiving agent. Whether noted or concealed, damage **MUST** be reported to the carrier **immediately** upon discovery, or in any event, within **14** days after receipt, and the contents and containers held for inspection by the carrier. A transportation company will not pay a claim for damage if an inspection is not requested within this **14** day period.

File a report with

- Call 1–800–548–3366 and use option 8.
- Contact your local service coordinator for more information on this process.

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WARNING

- THIS SERVICE MANUAL IS AVAILABLE IN ENGLISH ONLY.
- IF A CUSTOMER'S SERVICE PROVIDER REQUIRES A LANGUAGE OTHER THAN ENGLISH, IT IS THE CUSTOMER'S RESPONSIBILITY TO PROVIDE TRANSLATION SERVICES.
- DO NOT ATTEMPT TO SERVICE THE EQUIPMENT UNLESS THIS SERVICE MANUAL HAS BEEN CONSULTED AND IS UNDERSTOOD.
- FAILURE TO HEED THIS WARNING MAY RESULT IN INJURY TO THE SERVICE PROVIDER, OPERATOR OR PATIENT FROM ELECTRIC SHOCK, MECHANICAL OR OTHER HAZARDS.

AVERTISSEMENT

- CE MANUEL DE MAINTENANCE N'EST DISPONIBLE QU'EN ANGLAIS.
- SI LE TECHNICIEN DU CLIENT A BESOIN DE CE MANUEL DANS UNE AUTRE LANGUE QUE L'ANGLAIS, C'EST AU CLIENT QU'IL INCOMBE DE LE FAIRE TRADUIRE.
- NE PAS TENTER D'INTERVENTION SUR LES ÉQUIPEMENTS TANT QUE LE MANUEL SERVICE N'A PAS ÉTÉ CONSULTÉ ET COMPRIS.
- LE NON-RESPECT DE CET AVERTISSEMENT PEUT ENTRAÎNER CHEZ LE TECHNICIEN, L'OPÉRATEUR OU LE PATIENT DES BLESSURES DUES À DES DANGERS ÉLECTRIQUES, MÉCANIQUES OU AUTRES.

WARNUNG

- DIESES KUNDENDIENST-HANDBUCH EXISTIERT NUR IN ENGLISCHER SPRACHE.
- FALLS EIN FREMDER KUNDENDIENST EINE ANDERE SPRACHE BENÖTIGT, IST ES AUFGABE DES KUNDEN FÜR EINE ENTSPRECHENDE ÜBERSETZUNG ZU SORGEN.
- VERSUCHEN SIE NICHT, DAS GERÄT ZU REPARIEREN, BEVOR DIESES KUNDENDIENST-HANDBUCH ZU RATE GEZOGEN UND VERSTANDEN WURDE.
- WIRD DIESE WARNUNG NICHT BEACHTET, SO KANN ES ZU VERLETZUNGEN DES KUNDENDIENSTTECHNIKERS, DES BEDIENERS ODER DES PATIENTEN DURCH ELEKTRISCHE SCHLÄGE, MECHANISCHE ODER SONSTIGE GEFAHREN KOMMEN.

AVISO

- ESTE MANUAL DE SERVICIO SÓLO EXISTE EN INGLÉS.
- SI ALGÚN PROVEEDOR DE SERVICIOS AJENO A GEMS SOLICITA UN IDIOMA QUE NO SEA EL INGLÉS, ES RESPONSABILIDAD DEL CLIENTE OFRECER UN SERVICIO DE TRADUCCIÓN.
- NO SE DEBERÁ DAR SERVICIO TÉCNICO AL EQUIPO, SIN HABER CONSULTADO Y COMPRENDIDO ESTE MANUAL DE SERVICIO.
- LA NO OBSERVANCIA DEL PRESENTE AVISO PUEDE DAR LUGAR A QUE EL PROVEEDOR DE SERVICIOS, EL OPERADOR O EL PACIENTE SUFRAN LESIONES PROVOCADAS POR CAUSAS ELÉCTRICAS, MECÁNICAS O DE OTRA NATURALEZA.

ATENÇÃO

- ESTE MANUAL DE ASSISTÊNCIA TÉCNICA SÓ SE ENCONTRA DISPONÍVEL EM INGLÊS.
- SE QUALQUER OUTRO SERVIÇO DE ASSISTÊNCIA TÉCNICA, QUE NÃO A GEMS, SOLICITAR ESTES MANUAIS NOUTRO IDIOMA, É DA RESPONSABILIDADE DO CLIENTE FORNECER OS SERVIÇOS DE TRADUÇÃO.
- NÃO TENHA TENTADO REPARAR O EQUIPAMENTO SEM TER CONSULTADO E COMPREENDIDO ESTE MANUAL DE ASSISTÊNCIA TÉCNICA.
- O NÃO CUMPRIMENTO DESTA AVISO PODE POR EM PERIGO A SEGURANÇA DO TÉCNICO, OPERADOR OU PACIENTE DEVIDO A CHOQUES ELÉTRICOS, MECÂNICOS OU OUTROS.

AVVERTENZA

- IL PRESENTE MANUALE DI MANUTENZIONE È DISPONIBILE SOLTANTO IN INGLESE.
- SE UN ADDETTO ALLA MANUTENZIONE ESTERNO ALLA GEMS RICHIEDE IL MANUALE IN UNA LINGUA DIVERSA, IL CLIENTE È TENUTO A PROVVEDERE DIRETTAMENTE ALLA TRADUZIONE.
- SI PROCEDA ALLA MANUTENZIONE DELL'APPARECCHIATURA SOLO DOPO AVER CONSULTATO IL PRESENTE MANUALE ED AVERNE COMPRESO IL CONTENUTO.
- NON TENERE CONTO DELLA PRESENTE AVVERTENZA POTREBBE FAR COMPIERE OPERAZIONI DA CUI DERIVINO LESIONI ALL'ADDETTO ALLA MANUTENZIONE, ALL'UTILIZZATORE ED AL PAZIENTE PER FOLGORAZIONE ELETTRICA, PER URTI MECCANICI OD ALTRI RISCHI.

警告

- ・ このサービスマニュアルには英語版しかありません。
- ・ GEMS以外でサービスを担当される業者が英語以外の言語を要求される場合、翻訳作業はその業者の責任で行うものとさせていただきます。
- ・ このサービスマニュアルを熟読し理解せずに、装置のサービスを行わないで下さい。
- ・ この警告に従わない場合、サービスを担当される方、操作員あるいは患者さんが、感電や機械的又はその他の危険により負傷する可能性があります。

注意:

- 本维修手册仅存有英文本。
- 非 GEMS 公司的维修员要求非英文本的维修手册时，客户需自行负责翻译。
- 未详细阅读和完全了解本手册之前，不得进行维修。
- 忽略本注意事项会对维修员，操作员或病人造成触电，机械伤害或其他伤害。

Revision History

Date	Rev	Change Description
14-Feb-2006	1	Initial Release
14-Jul-2006	2	Section 1: Removed information that a second MUX board was to be installed. Section 2: Removed instructions in for removing and installing MUX board in slots 13 and 14. (MRIhc16250) Added step for connecting cable to J5 on MUX board.
10-Jun-2009	3	Section 2: Added Mux board 5250106 to step 2 in Section 2-2 (PQR 13238441)

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SECTION 1 – OVERVIEW

1-1 SIGNA HORIZON SYSTEM CONFIGURATION

This manual provides instructions to upgrade a Signa EXCITE HD or HDx system to a 16 Channel system.

This Direction covers the following catalogs:

M3000PE – 16 Channel RF Electronics Upgrade

This upgrade adds:

- RF-DIF, MUX, UTNS3, and RCVR II boards.
- Cables

1-2 SITE READY CHECK

Pre-installation work must be completed before equipment is delivered to avoid delays and confusion.

1-3 DISCARDED MATERIAL RETURN POLICY

The equipment or material being removed by the upgrade have a potential value for service of the installed base. The disposition of such must be controlled by GE. Disposition of the removed material is per GE Medical Systems Policy and Procedures.

1-4 PRODUCT LOCATOR

The Global Install Base Database (also known as Product Locator System) tracks shipment, trans-shipment, and field location of the serialized models. There are now two methods for submitting Product Locator information.

1. At this time, for **U.S. ONLY**, the preferred method of submitting information is the *FE Site Verification Web Site* at:
<http://gein2.med.ge.com/gib>

The FE Site Verification consists of three components that are available on the web from the main menu. They are:

- Install/deinstall product locator model and serial numbers
- Add/modify ship to address information
- Update CARES FE data for primary/secondary FEs

To obtain a copy of the FE training tutorial for using this website, a downloadable copy is available at the following:
Product Locator Support Central Page – <http://supportcentral.ge.com/15563>

2. One "Shipping Card" is filled out and submitted when shipped (extra cards are supplied for trans-shipments between storage and distribution points), and the "Installation Card" and extra shipment cards are attached.

Verify that serial and model number on each rating plate matches installation card numbers before removing installation card. Note that there may be one or more shipment cards and bar code labels with the installation card. These shipment cards are used to trace the transfer of serialized units between various inventory storage and distribution points until the product reaches its final installation destination. Process just the installation card and discard any extra shipment cards and labels.

1-5 SOFTWARE LOAD FOR UPGRADE

Install 16 Channel option key.

1-6 FINAL COORDINATION

1. Replace all covers that have not been previously replaced.
2. Record and enter applicable data into applicable site configuration files and records.
3. Process product locator installation cards for all serialized components added to the system to include product locator card supplied with the upgrade serial number rating plate. Return to:

**Product Locator File
P.O. Box 414, W-523
Milwaukee, WI 53201-0414**

Note

Failure to fill out and return Product Locator Cards may result in failure of your site to receive future FMI's.

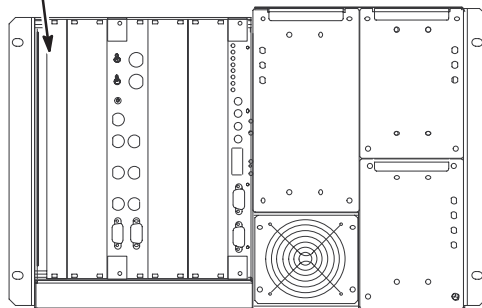
SECTION 2 – UPGRADE PROCEDURE

2-1 REMOVE EXISTING COMPONENTS FROM RRF CHASSIS



Note: Module is a static sensitive component and needs to be handled as such (wrist strap).

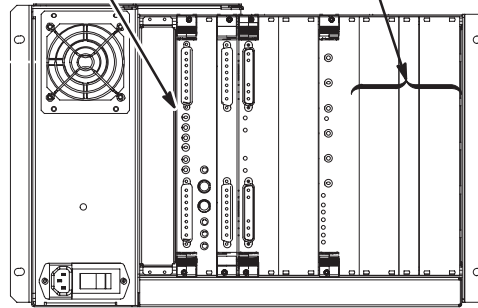
- ① Remove blank panel from slot 1 at front of chassis.



FRONT VIEW

Note: It is recommended that the existing MUX board be at the same part number and revision level as the new MUX board being installed in slots 4 & 5.

- ② Remove blank panels from slots 1 to 5 at rear of chassis.

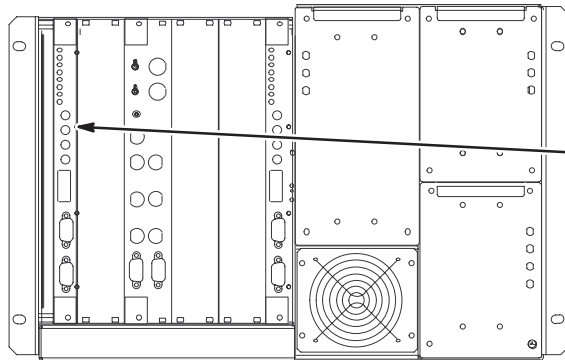


REAR VIEW

2-2 INSTALL MODULES IN RRF CHASSIS

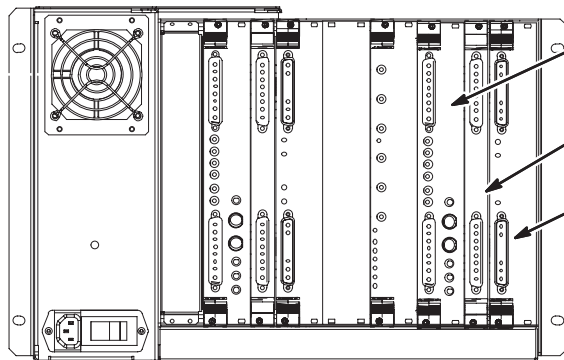


Note: Module is a static sensitive component and needs to be handled as such (wrist strap).



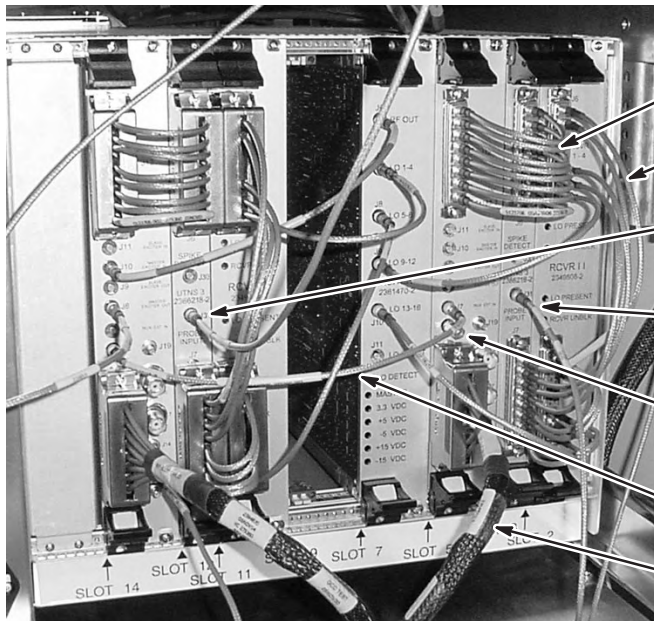
FRONT VIEW

- ① Install RF-DIF Board (5250028) in slot 1.



REAR VIEW

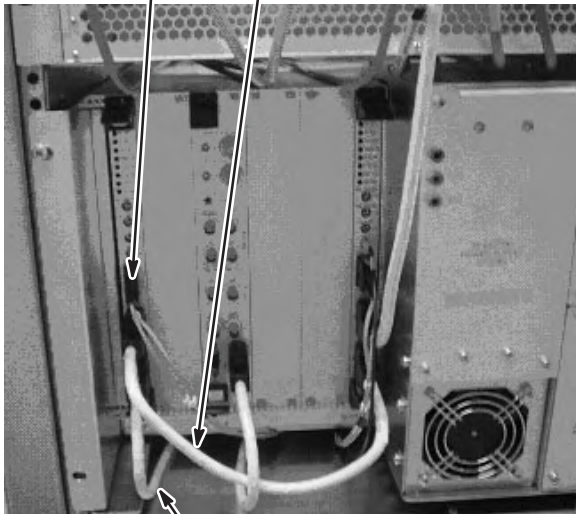
- ② Install MUX Board (2274188-3 or 5250106) in slots 4 & 5 .
- ③ Install UTNS3 Board (2366218-2) in slot3.
- ④ Install RCVR II Board (2349808-2) in slots 1 and 2.

2-3 INSTALL CABLES AT REAR OF RRF MODULE

- ① Connect cable 5123706 as marked.
- ② Connect cable 5113193-2 as marked.
- ③ Make sure cable 5115417 slot 12R J31 is connected as marked. it should be routed and connect to Coax Power Splitter "1".
- ④ Connect cable 5115417-2 as marked. Route and connect to Coax Power Splitter "2". Remove 50 Ohm terminator if installed at "2".
- ⑤ Move 50 Ohm Terminator from Mux1 J6 (slots 13 & 14) to Mux2 J6 (slots 4 & 5).
- ⑥ Connect cable 5115630 as marked.
- ⑦ Connect cable 5145030-2 (HDx) or 2394870-2 (HD) to new MUX board.

2-4 INSTALL CABLES AT FRONT OF RRF MODULE

- ③ Connect cable 5121122-2 as marked to RF Slot 1F J21. Route and connect other end to MGD Slot 11F J7.
- ② Connect cable 5113182-2 as marked.
- ① Disconnect cable (2323878-3) from Slot 10F J23 and connect to Slot 1F J23.



2-5 INSTALL NEW REAR DOOR AND RATING PLATE

- ① Attach upgrade Rating Plate (46-302200P12) to top of cabinet on back side.

